Affect- and Uncertainty-Aware Adaptive Multi-Agent Clarification for Multilingual Emergency Calls

Abstract*--Emergency call-taking in Sri Lanka has to be robust to speech in Sinhala, Tamil, English and frequent code-mixing under challenging acoustic conditions. Single-pass speech understanding is often insufficient due to degraded initial utterances. We propose a closed-loop spoken dialogue system that incrementally converts multilingual calls to dispatch reports by asking a single missing information item at a time to clarify. Six cooperating agents work together across domains including speech recognition, language and code-mix identification, relevance-gated entity extraction, protocol-aligned readiness assessment, adaptive clarification and constrained realization with safety validation, and language-routed speech synthesis. Our major contribution is a novel explicit affect- and uncertainty-aware finite-state clarification policy that is auditable and bounded rather than relying on a black-box generative model. Evaluated on 510 simulated multilingual emergency calls, entity F1 improves from 0.278 to 0.429 and relevance F1 from 0.783 to 0.819. Protocol-aware readiness reduces critically incomplete classifications from 91.4% to 74.1%; clarification diversity increases from 0.040 to 0.222 with 100% safety pass-rate; and spoken-response language match from 41.0% to 100%.

***Keywords*--emergency call-taking, multilingual NLP, code-mixing, dialogue management, clarification policy**

Introduction

Emergency call-taking is an information gathering task under severe time pressure. The call-taker has to establish, where the incident is, what is happening, how many people are involved, and whether anyone is injured or trapped while the caller may be frightened, breathless or talking from a noisy scene. Delays in resolving ambiguities can delay dispatch.

Sri Lanka introduces additional multi-lingual challenges to the conversation understanding task. Both Sinhala and Tamil are recognized as official languages in the country and English is used widely as a lingua franca; callers may also mix and match the languages within one utterance. For instance, a call can start with a question in Sinhalese, contain an English-coded landmark description and end with details given in Tamil. Such code-switched utterances pose significant challenges to automatic speech recognition systems trained to work with one language only and downstream information extraction suffers from transcription errors.

Transcription of the report and extraction of information in this way is therefore not enough. The system must operate by determining whether the accumulated report is sufficient for dispatch or what single question is most urgent to ask next to fill in the critical gaps. In other words, a dialogue is needed within strict safety constraints: no guessing, no asking more than one question at a time, no giving advice, no assuming a "no" when there is a long pause, and no getting stuck in a loop.

This paper contributes the following.

- cooperative multi-agent decomposition which partitions perception, assessment, decision, realization and delivery, granting disjoint authority over each component
- relevance-gated multilingual extraction stage which prevents off-topic or non-informative utterances from polluting the structured report
- protocol-aligned, confidence-aware dispatch-readiness assessment which scores proximity to dispatch-readiness rather than binary null-checks
- affect- and uncertainty-aware clarification policy which selects among 7 strategies, adapts to detected distress, treats silence as bounded channel failure and limits redirections
- language-routed speech synthesis with explicit safety and style constraints, validated against full closed-loop pipeline on a 510-call multilingual corpus

I. Related Work

A. Multilingual and Code-Mixed Understanding

Cross-lingual pretrained encoders such as XLM-RoBERTa [1] provide shared representations across many languages and can transfer to low-resource targets with limited supervision. Large-scale weakly supervised speech models [2] similarly broaden multilingual recognition, although intra-sentential code-mixing remains difficult. Prior work on code-switched language identification [3] motivates treating mixed varieties as explicit classes; accordingly, this work models Singlish and Tanglish directly rather than as noisy monolingual speech.

B. Sequence Labelling for Information Extraction

Conditional random fields [4] constitute a strong feature-based baseline in the task of span extraction, while bidirectional LSTM-CRF architectures [5], [6] were the dominant neural approach prior to the advent of transformer fine-tuning. Both are included as baselines here to measure the benefit of pretraining on the small domain corpus across multiple languages.

C. Multi-Agent and Task-Oriented Dialogue

Task-oriented dialogue systems persist state over slot values and select subsequent system actions [7], [8]. Modular approaches decompose the problem into perception, state tracking, policy, and generation, at the expense of joint optimization, and enable more inspectable policies. The literature on clarification questioning typically ranks candidate questions by expected information gain [9], [10]. Emergency call-taker systems differ in that the slot inventory is largely determined by protocol, there is a strong cost associated with taking too long to reach an actionable state, and the user may be in distress; these are factors that motivate an explicit, inspectable policy over a generative one.

D. Emergency Protocols and Alerting Standards

Emergency call-handling guidance [11], [12] focuses on classes of information such as where, what, who, when, and life safety, along with a minimum set of information that must

be obtained before resources can be committed. The Common Alerting Protocol [13] adds urgency, severity, and certainty. The readiness agent is engineered to these concepts; it is not a certified implementation of either standard.

E. Multilingual Speech Synthesis

Neural speech synthesis supports many languages [14], [15], but the availability and quality of voices varies for Sinhala and Tamil. An English voice reading non-English text would render a clarification unusable. Consequently, the synthesis stage directs each detected language class to an engine-voice pair that has been verified for that script.

II. System Architecture

A. Turn Loop and Session State

Fig. 1 shows the closed loop executed for each call turn. The audio is passed in at the top, and a clarifying utterance is emitted at the bottom as the prompt to the next one. The loop terminates when the readiness agent announces DISPATCH_READY and the accumulated report is serialized, or by a terminal action issued by the policy agent.

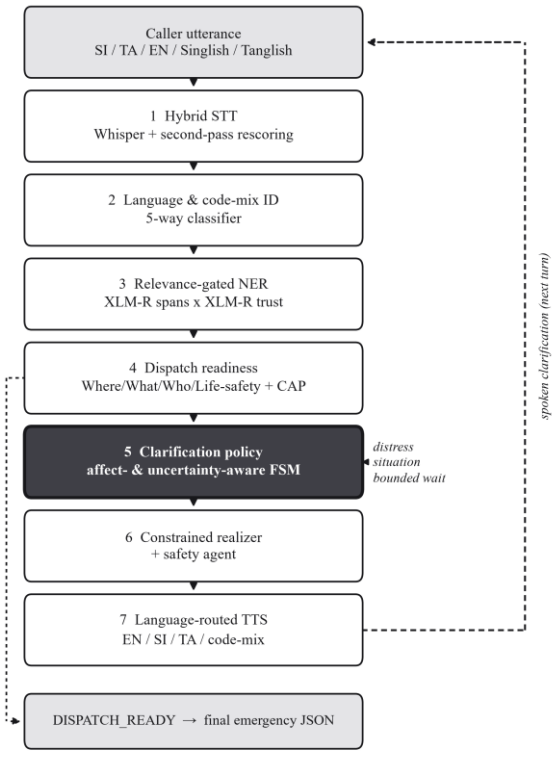


Fig. 1. Closed-loop turn architecture. Stage 5, the research contribution, takes in fused dialogue state along with distress signals, situation models, and bounded-wait constraints, and outputs a single clarification strategy.

Session state is maintained across turns. Newly trusted spans are merged into the consolidated report, without overwriting stronger evidence, thus avoiding later unintelligible turns. A value is only replaced with another one if the corresponding field is empty, or the new evidence is more structured than the previous one (e.g., exact address instead of area). Field-level timestamps are used to capture the chronological order of information acquisition.

B. Agent Decomposition

The six agents have disjoint decision authority, perception agent takes care of recognition, language identification and span extraction. Trust agent decides if utterance is relevant can veto extracted spans. Readiness agent will score the slots and nominate one target field; affect agent estimates distress and situation signals. Policy agent is the only one which selects an action, while realization and safety agent translates the action into one validated sentence and can also veto unsafe wording.

This separation renders each turn individually auditable since the policy agent cannot generate text, realization agent cannot select the action. Therefore, a question which reaches the caller has an explicit strategy, target, justification, an independent safety.

III. Methodology

A. Hybrid Multilingual Speech Recognition

Recognition works by candidate generation and selection. The audio is normalized to 16khz mono, language hint is estimated, and a forced-language whisper pass generates a transcript with average token log-probability. Low confidence or very short clips are sent to a second recognizer. Persistently degraded candidates are re-decoded under alternate language codes, jointly scored, and passed through a domain-specific refinement step, which reduces the risk of accepting a confidently decoded transcript in an incorrect language.

B. Language and Code-Mix Identification

A character- and word-based classifier was used to assign each transcript to either en-US, si-LK, ta-IN, si-en, or ta-en. The mixed classes were learned directly by the classifier as they differ from the monolingual variants in downstream question wording, language validation, and synthesis routing.

C. Relevance-Gated Emergency Entity Extraction

Extraction uses two fine-tuned multilingual encoders. A token classifier predicts seven entity types: ADDRESS_EXACT, ADDRESS_AREA, PLACE_TYPE, INCIDENT, PEOPLE_COUNT, INJURY, and TRAPPED. A second model predicts utterance relevance in {EMERGENCY_RELEVANT, CLARIFICATION_RESPONSE, IRRELEVANT, NON_INFORMATIVE} and dialogue act in {PROVIDE_INFO, CONFIRM, DENY, ASK_REPEAT, DISTRESS, OTHER}.

Relevance gates extraction: spans from IRRELEVANT or NON_INFORMATIVE utterances are not merged into the report. Dialogue act also conditions the policy because DENY and ASK_REPEAT require different responses from PROVIDE_INFO even when entity spans are similar.

Sinhala and Tamil person counts receive dedicated numeric normalization. A compositional lexicon converts spoken compounds to integers before the count slot is scored, distinguishing definite quantities from vague expressions.

D. Protocol-Aligned Dispatch Readiness

Extracted slots are associated with WHERE_LOCATION, WHAT_INCIDENT, WHO_PEOPLE, LIFE_SAFETY_INJURY, LIFE_SAFETY_TRAPPED and a contextual WHEN_CONTEXT category. Each of the slots was labeled READY, WEAK or MISSING using structural sufficiency, extractor confidence, source-utterance trust, and corroborating fields. Type-specific rules favor exact address

and normalized integer counts over area names and hedged quantities.

Overall readiness aggregates the per-slot completeness scores against their dispatch weights,

$$P = (\copyright\ \omega_\iota\ \chi_\iota) / \copyright\ \omega_\iota \qquad (1)$$

where c_i is the completeness of slot i and w_i its dispatch weight, with location weighted highest and time context lowest. Readiness is discretized into {DISPATCH_READY, CLARIFICATION_REQUIRED, NOT_READY_CRITICAL}. Urgency, severity, and certainty are inferred in parallel from life-safety and relevance signals, and a single clarification target is nominated by the agent based on protocol priority and gap severity.

E. Affect- and Uncertainty-Aware Clarification Policy

The policy agent is an explicit, non-generative decision maker. It takes dialogue state, distress, and situation assessments as inputs and outputs one strategy, one target field, and a machine-parsable justification. Fig. 2 shows the decision-making order.

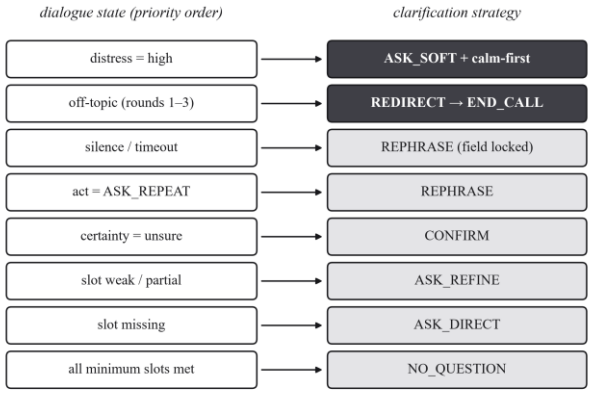


Fig. 2. Clarification strategy selection. Conditions are evaluated in the order shown, so affect and dialogue-level failures preempt slot-level gaps.

Distress detection involves the merging of the upstream signals rather than relying on language-specific emotion terms. The rating takes into account dialogue act, relevance, urgency and severity calculations, and acoustic cues,

$$\delta = \copyright \llcorner_\kappa \phi_\kappa, \quad \chi\alpha\lambda\mu_\phi\iota\rho\sigma\tau = [\ \delta\ \varepsilon\ |_{\eta\iota\gamma\eta}\ ] \qquad (2)$$

and is thresholder into none, mild, or high. High distress will set a calm-first flag and pick an ASK_SOFT (with additional reassurance before the ask).

Repeated out-of-topic behavior is penalized with finite adaptive termination: first incident redirects softly, second incident redirects more firmly, third incident terminates or hands off after a final warning. An explicit confirmation of no emergency terminates the call, and the system does not re-open the gate after handoff.

Silence is treated as a channel failure (rather than refusal). After every ask there is a bounded wait budget composed of a base delay, field weight, and recent pipeline latency. Timeouts lock the current target and either route to REPHRASE or, if distressed, to ASK_SOFT. A generation counter ensures that timeouts from previous, superseded questions are not mistaken for out-of-topic or no-emergency termination.

F. Constrained Realization and Safety Validation

The selected strategy ensures the use of templates with control from the caller language and can be based on an already-given span. The safety agent makes sure that the output contains only one bounded-length question, targets a protocol field, matches the detected language, and does not contain medical or tactical advice nor repeats the previous utterance verbatim. All failed outputs default to the validation of a template for the field and strategy.

G. Language-Routed Emergency Speech Synthesis

The speech synthesis path depends on the detected language and proceeds to the validated engine-voice pair. The languages include a neural commercial voice for English, a neural local voice for Sinhala, and a neural regional voice for Tamil. For Singlish and Tanglish, the respective Sinhala or Tamil path is used, and embedded English words are synthesized. Voices in English are not allowed in Sinhala, Tamil, and mixed languages.

IV. Experimental Setup

A. Corpus

The dataset comprises 510 artificially-generated single-channel calls to emergency services dealing with fires, road accidents, medical issues, floods, and violent situations and is divided into five language categories. There are 502 calls that have stage outputs and this makes the batch set.

B. Annotation and Splits

For entity recognition task, 306 utterances were manually annotated, resulting in 783 gold spans. Character-overlap projection resulted in 773 aligned spans (hit rate is 0.9872). Fixed seed split of 212/43/51 was used, ensuring language distribution; the test set has 101 entities.

Relevance and dialogue act classification tasks consist of labeling of 212 utterances that were split into 142/30/40. The dataset consists of 178 manually labeled and 34 artificially generated instances to balance rare classes; test split has 33 manual and 7 artificial utterances.

C. Baselines and Metrics

Entity recognition is compared to CRF and BiLSTM-CRF benchmarks by means of exact-span micro precision, recall, F1, and token accuracy scores. Relevance and dialogue act classification tasks are compared to TF-IDF logistic regression and SVM classifiers through weighted F1 score and accuracy scores. The readiness task is compared to the baseline of performing a null check. The clarification task is compared to the baseline of using fixed per-field templates through question diversity and safety pass rates.

D. Implementation

Both models are based on xlm-roberta-base. The entity model is trained for 10 epochs with a learning rate of 3×10^{-5}, while the relevance model is trained for 8 epochs and the best performing checkpoint is selected on the development set in terms of relevance F1 (0.8623). The python runtime loads the models lazily and exposes browser-microphone and telephony channels over the same pipeline.

V. Results and Discussion

A. Emergency Entity Recognition

Comparison among extractors in the held-out test set is provided in Table I. XLM-R performs better than the two baselines, achieving an improvement in F1 of 15.1 over the superior baseline.

TABLE I. Emergency entity recognition on the test split

Model	P	R	F1	Acc.
CRF (baseline)	0.373	0.218	0.275	0.624
BiLSTM-CRF (baseline)	0.400	0.213	0.278	0.716
XLM-R (proposed)	0.500	0.376	0.429	0.724

The gain is mostly due to recall, where precision increases from 0.400 to 0.500 and recall from 0.213 to 0.376. In our context, failure to identify a span results in the necessity for an extra explanation and thus an additional delay.

As demonstrated in Table II, there is considerable variance for each entity. The best-performing ones are PEOPLE_COUNT and TRAPPED, whereas the worst ones get 0.

TABLE II. Per-entity results of the proposed extractor

Entity	P	R	F1	Sup.
ADDRESS_AREA	0.500	0.143	0.222	7
ADDRESS_EXACT	0.500	0.389	0.438	18
INCIDENT	0.476	0.400	0.435	50
INJURY	0.000	0.000	0.000	2
PEOPLE_COUNT	0.714	0.833	0.769	6
PLACE_TYPE	0.000	0.000	0.000	7
TRAPPED	1.000	0.455	0.625	11
Micro average	0.500	0.376	0.429	101

INJURY only gets four training references and two test references, which does not give enough information for learning. PLACE_TYPE is mostly part of larger spans, losing the contest of boundaries. Later stages of readiness and policies help with this by asking for the field that needs attention.

Table III provides the results per language class. Singlish shows the best performance, as expected due to the Latin alphabet being more easily identifiable than Sinhala or Tamil.

TABLE III. Per-language F1 of the proposed extractor

Language class	F1	Test sent.
si-en (Singlish)	0.577	11
si-LK (Sinhala)	0.471	12
en-US (English)	0.407	15
ta-IN (Tamil)	0.235	7
ta-en (Tanglish)	0.200	6

Tamil proves to be least effective in terms of monolingualism and hybridism. It accounts for only 52 out of 212 training sentences and thus has the smallest share among languages.

B. Utterance Relevance and Dialogue Act

Table IV shows the result of gating classification. XLM-R outperforms two baselines in relevance and achieves 0.761 weighted F1 on dialogue act.

TABLE IV. Utterance relevance and dialogue act

Model	F1	Acc.
Logistic regression (base.)	0.783	0.800
SVM (baseline)	0.765	0.800
XLM-R (proposed)	0.819	0.825
XLM-R, dialogue act	0.761	0.800

CLARIFICATION_RESPONSE achieves 1.000 F1, while EMERGENCY_RELEVANT achieves 0.910. Errors tend to be for NON_INFORMATIVE (0.550 F1, recall 0.43), meaning that some empty turns will go through the filter. Rare dialogue acts cannot be reliably classified: there are only two test examples for each of DISTRESS, CONFIRM, and DENY. The model, consequently, combines several features, not just the dialogue act one.

The language ID achieves 0.889 accuracy and 0.878 macro F1. English is completely separated, while the hardest case is the Sinhala-English mix, which shares vocabulary with both adjacent languages, achieving 0.743 F1.

C. Dispatch Readiness

Comparison between readiness assessment for 502 calls is provided in Table V. In the case of the proposed approach, an agent is used which is not trained but rule-based.

TABLE V. Dispatch readiness over 502 calls

Method	Ready	Critical
Null/empty field check	8.6% (43)	91.4% (459)
Protocol conf.-aware	8.4% (42)	74.1% (372)

In terms of the number of calls in the ready set (42 versus 43 calls), the two approaches are close. However, they differ significantly in the number of calls marked as critically important (372 versus 459). The null approach marks all instances where there is no data in the required fields as critically incomplete.

D. Clarification Policy

Table VI reveals that diversity in question rises from 0.040 to 0.222 (5.6 times) and that all questions produced are safe and have appropriate language tags.

TABLE VI. Clarification behaviour over 502 calls

Measure	Fixed tmpl.	Proposed
Question diversity	0.040	0.222
Safety pass rate	—	1.000
Language tag rate	—	1.000
No-question rate	—	0.084

The policy generated 281 ASK_DIRECT, 97 ASK_SOFT, 82 ASK_REFINE, and 42 NO_QUESTION decisions. Location is an example of how the policy adjusted: 249 calls got direct asks, 82 calls got refine asks because of existing poor location data, and 74 calls got soft asks owing to distress. Consequently, one missing attribute may result in a variety of approaches depending on the state.

An 8.4% no-question ratio reflects calls where the policy decided not to ask because of readiness or non-emergency.

E. Multilingual Speech Synthesis

Table VII assesses 39 clarification questions sampled. All of them use language routing correctly, whereas in the baseline (only English), 41.0% are correct.

TABLE VII. CLARIFICATION SPEECH SYNTHESIS

Measure	English base.	Proposed
Language match rate	0.410	1.000
Style constraint pass	—	1.000
Blocked utterances	—	0.000

The samples have 16 English, 10 Sinhalese, and 13 Tamil queries. Language match is the criterion for routing accuracy and not the perceived quality of the audio. The end-to-end test was successful in the case of non-English synthesis with latency values of 1.20 s and 1.40 s for Sinhala and 1.18 s for Tamil queries.

F. Overall Comparison

Fig. 3 compares four modules with their baselines on a common higher-is-better scale.

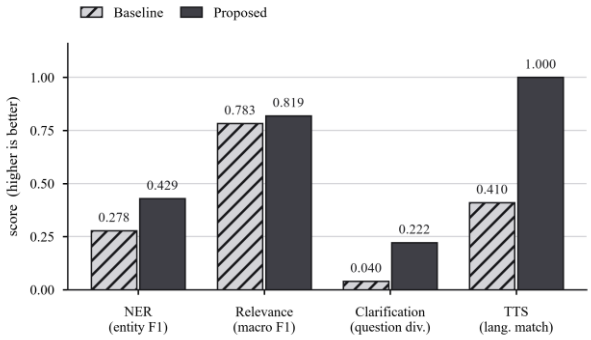


Fig. 3. Baseline against proposed across four pipeline modules. All four measures are oriented so that higher is better.

All four components benefit, albeit with varying amounts. Relevance improves by 3.6 points as a result of its strong TF-IDF baseline, while extraction improves by 15.1 points; clarification and synthesis get more improvement because they are no longer hard-coded. Most significantly, extraction mistakes are not overlooked.

G. Limitations

Limitations of the study include three major factors. Firstly, the dataset used in this work is artificial; thus, the actual emergency sounds and behaviors might be more diverse. Secondly, annotated datasets for testing are limited in number, particularly for INJURY, DISTRESS, and DENY. Lastly, assessment of speech synthesis is based on routing accuracy and adherence to the style rather than on native speaker comprehensibility.

VI. CONCLUSION

This paper introduced a closed-loop multilingual emergency call system that modeled the process of clarification as an explicit decision-making problem involving six agents. Rather than extraction once and the missing field report, the system continuously evaluated dispatch readiness and selected a single clarification strategy based on a fused state of caller affect, dialogue history, and silence.

With 510 synthetic Sinhala, Tamil, English, Singlish, and Tanglish calls, multilingual fine-tuning boosted entity F1 from 0.278 to 0.429 and relevance F1 from 0.783 to 0.819. Protocol-aware readiness improved critical classification from 91.4% to 74.1%; adaptive clarification augmented diversity from 0.040 to 0.222 with a 100% safety pass rate; and language-routing synthesis improved response language matching from 41.0% to 100%.

The primary novelty is the separation of an explicit decision-making policy from the constrained execution of responses. Every question contains its strategy, target, and rationale, and one question per output, no advice, waiting limit, and redirection limit constraints are imposed in structural form. Auditability of this kind is necessary in future emergency line systems where the reasoning behind the clarification is crucial.

REFERENCES

[1] A. Conneau et al., "Unsupervised cross-lingual representation learning at scale," in Proc. 58th Annu. Meeting Assoc. Comput. Linguistics, 2020, pp. 8440–8451.

[2] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in Proc. 40th Int. Conf. Mach. Learn., 2023, pp. 28492–28518.

[3] T. Solorio et al., "Overview for the first shared task on language identification in code-switched data," in Proc. 1st Workshop Comput. Approaches Code Switching, 2014, pp. 62–72.

[4] J. Lafferty, A. McCallum, and F. C. N. Pereira, "Conditional random fields: Probabilistic models for segmenting and labeling sequence data," in Proc. 18th Int. Conf. Mach. Learn., 2001, pp. 282–289.

[5] G. Lample, M. Ballesteros, S. Subramanian, K. Kawakami, and C. Dyer, "Neural architectures for named entity recognition," in Proc. NAACL-HLT, 2016, pp. 260–270.

[6] Z. Huang, W. Xu, and K. Yu, "Bidirectional LSTM-CRF models for sequence tagging," 2015, arXiv:1508.01991.

[7] M. Henderson, B. Thomson, and J. D. Williams, "The second dialog state tracking challenge," in Proc. 15th Annu. Meeting Special Interest Group Discourse Dialogue, 2014, pp. 263–272.

[8] T. Zhao and M. Eskenazi, "Towards end-to-end learning for dialog state tracking and management using deep reinforcement learning," in Proc. 17th Annu. Meeting Special Interest Group Discourse Dialogue, 2016, pp. 1–10.

[9] S. Rao and H. Daumé III, "Learning to ask good questions: Ranking clarification questions using neural expected value of perfect information," in Proc. 56th Annu. Meeting Assoc. Comput. Linguistics, 2018, pp. 2737–2746.

[10] M. Aliannejadi, H. Zamani, F. Crestani, and W. B. Croft, "Asking clarifying questions in open-domain information-seeking conversations," in Proc. 42nd Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval, 2019, pp. 475–484.

[11] European Emergency Number Association, "112 operational requirements," EENA Operations Document, Brussels, Belgium, 2019.

[12] National Emergency Number Association, "NENA standard for 9-1-1 call processing," NENA-STA-020, Alexandria, VA, USA, 2020.

[13] OASIS, "Common Alerting Protocol version 1.2," OASIS Standard, Jul. 2010.

[14] J. Shen et al., "Natural TTS synthesis by conditioning WaveNet on mel spectrogram predictions," in Proc. IEEE Int. Conf. Acoust., Speech Signal Process., 2018, pp. 4779–4783.

[15] V. Pratap et al., "Scaling speech technology to 1,000+ languages," J. Mach. Learn. Res., vol. 25, no. 97, pp. 1–52, 2024.